

IE232 mini-project: Modeling Predator-Prey Oscillations

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1 Introduction: the model

For this project, I chose to model the evolution of the number of prey and predators in an ecosystem. I based the model on data on lions and antelopes.

The time unit of the system is months. At each iteration, births and deaths are computed and the population is updated. Deaths/survivors are modeled with **binomial distributions**, while births are modeled with **poisson distributions**. The parameters of the distributions are functions of the variables below and the populations, they were chosen in an attempt to model real world dynamics between predator and prey populations.

2 Parameters

The **reproduction rate** have been chosen corresponding to real world data. I assumed that the proportion of males and females was the same. Lionesses give birth to an average of 2 offspring every 2 years. That's a rate of $(2/24) \times (1/2) + 0 \times (1/2) = 1/24$ per lion per month.

As for antelopes, females give birth to an average of 2 offspring per year, giving a rate of $(2/12) \times (1/2) = 1/12$ per antelope per month.

The **predation rate** corresponds to the number of antelopes hunted per lion per month, which is around 3.

The **k** variable was used to adjust the lions' survival probability so that it would be low if they ate few antelopes and high if they ate enough.

Carrying capacities were used to limit the proliferation of prey (due to limited natural resources/food). They were also used to regulate prey hunting according to prey and predator densities. The probability of an antelope being eaten is therefore given by:

$$p_{\text{hunted}} = \frac{\text{predation_rate} \times \text{predator_population}}{\text{prey_population}} \times \sqrt{\frac{\text{prey_population}}{\text{preys_carrying_capacity}}} \times \sqrt{\frac{1 - \text{predator_population}}{\text{predator_carrying_capacity}}}$$

Epsilon is used to prevent division by 0.

3 Results Analysis

The parameters values used to obtain the results are those of the notebook (unless specified).

The result of the simulation is an **oscillatory behavior** in the evolution of the number of predators and prey. This corresponds to the expected real-world behavior: if there is a lot of prey, predators survive and proliferate; if there is little prey, predators die, allowing prey to multiply again.

The periods of oscillation are variable and difficult to estimate, I found a period of about 10 years. The number of prey reaches a minimum of around 1100 and a maximum of 1550, and the number of predators a minimum of 10 and a maximum of 35. This can be explained by the parameters and formulas used. By increasing the carrying_capacity (from 2000 to 3000 for predators, from 50 to 100 for prey) we observe, for example, a maximum of 2200 for prey and 60 for predators. However, populations are much more versatile and extinction is more likely.

Population	Minimum	Maximum	Average	Std	Time Above Mean	Time Below Mean
Predators	9	37	20.99	4.57	621 months	580 months
Preys	1106	1535	1286.79	83.61	544 months	657 months

Table 1: Statistics for predator and prey populations computed over 100 years.

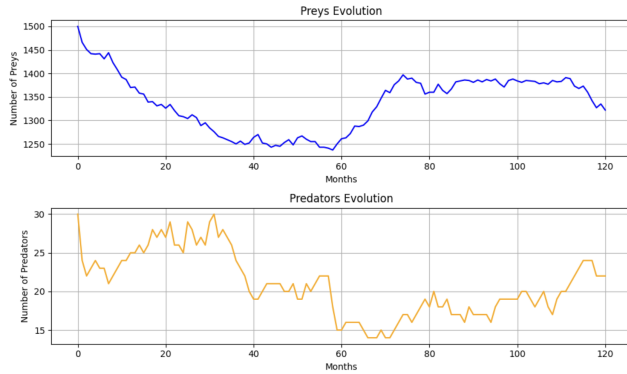


Figure 1: Population evolution over 10 years

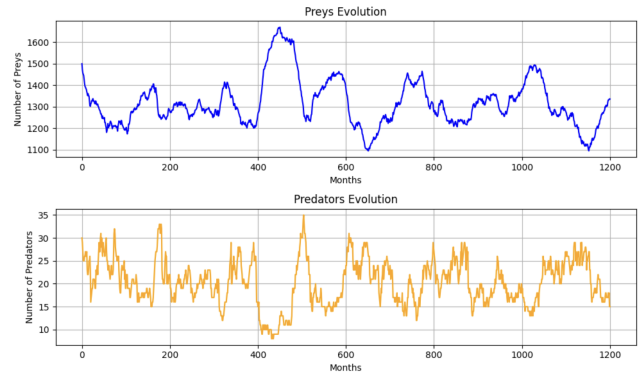


Figure 2: Population evolution over 100 years

In the long run, populations continue to oscillate.

4 Model - Limit cases

By setting a **low number of prey** (100) compared to the number of predators (30), the predators quickly all die from starvation and then the prey proliferate due to the absence of predators. This behavior may be close to reality, as predators have difficulty finding prey due to their rarity, but we could also think that prey become extinct, which doesn't happen in this model. If this is the desired behavior, we could modify the *p_hunted* formula so as to penalize low prey density less.

By setting a **low number of predators** (2) and a **large number of prey** (1500), 2 behaviors are possible. Either the predators won't succeed to reproduce and extinct, or their population will slowly increase until reaching oscillation.

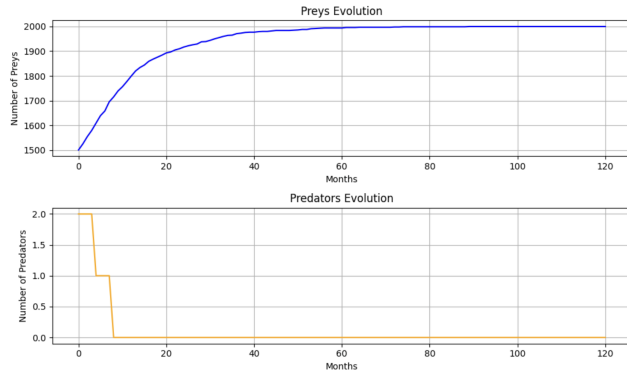


Figure 3: Evolution over 10 years with an initial low predator population, leading to predator extinction

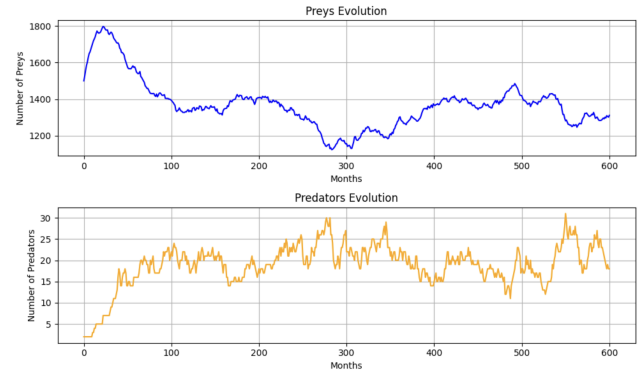


Figure 4: Evolution over 10 years with an initial low predator population, predators surviving

5 Possible Improvements

- In this model, animals reproduce on their own, for example if there is only one prey remaining, it can still reproduce. → Simulate each animal with a **gender**, females can be pregnant, babies born after a certain **gestation period**.
- Giving to individuals "**growth states**" → They can only reproduce after a certain age and are more likely to die as they get older.

In conclusion, the results of this work show the fragility of an ecosystem and the importance of preserving a stable ratio between predators and prey to ensure its sustainability.